

“Calculus 3”

Multi-Variable Calculus

Instructor: Álvaro Lozano-Robledo

Day 23

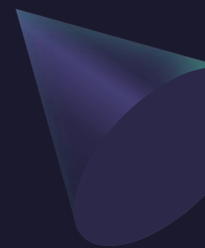


Any Reminders? Any Questions?

- Quiz on Friday (discussion) covers 16.3, 16.4 (fundamental theorem of line integrals and Green's theorem)
- Office hours Mondays 1-2pm online
- Office hours Thursday 3:30-4:30 at MONT 233
- Calc Night Thursday 6:30-8:30 at MONT 104



ALVARO: Start the recording!

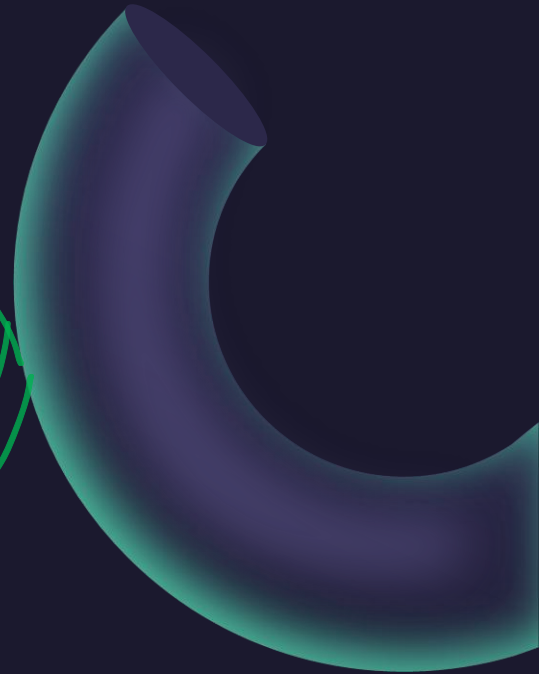
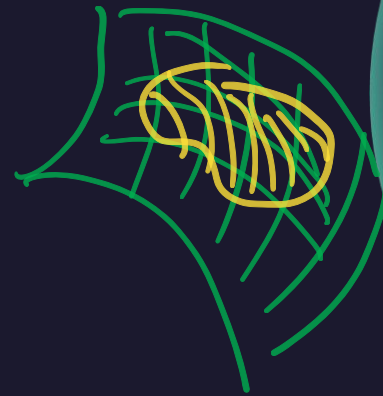
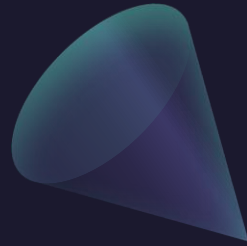


“Calculus 3”

Multi-Variable Calculus

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Parametric Surfaces

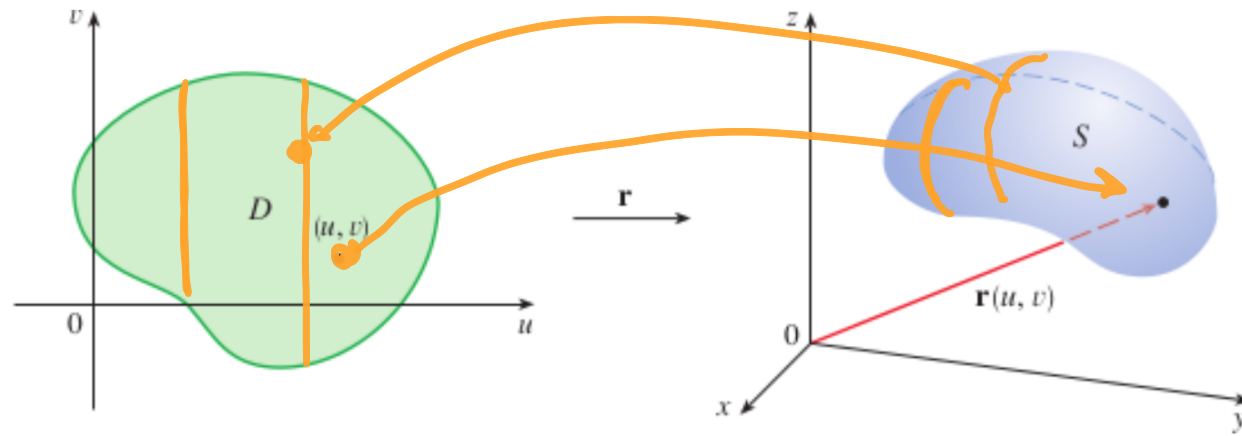


About Parametrized Surfaces

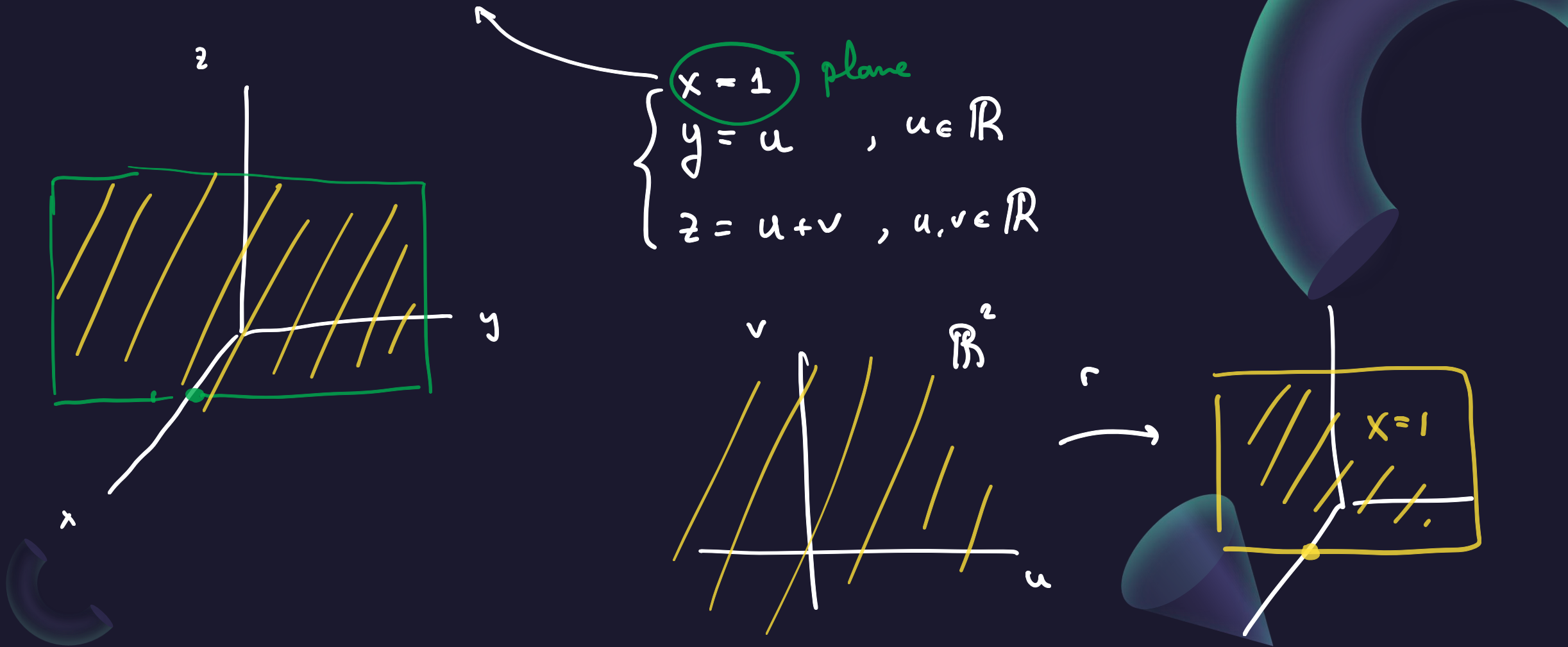
$$\mathbf{r}(t) = x(t)\mathbf{i} + y(t)\mathbf{j} + z(t)\mathbf{k} = (x, y, z)$$

$$\mathbf{r}(u, v) = x(u, v)\mathbf{i} + y(u, v)\mathbf{j} + z(u, v)\mathbf{k} = (x, y, z)$$

A parametric surface

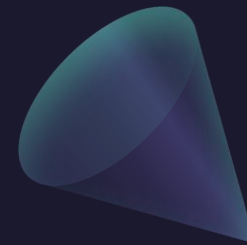


Example: Identify and sketch the surface with vector equation
$$r(u, v) = (1, u, u + v) = 1 \cdot i + u \cdot j + (u + v) \cdot k.$$

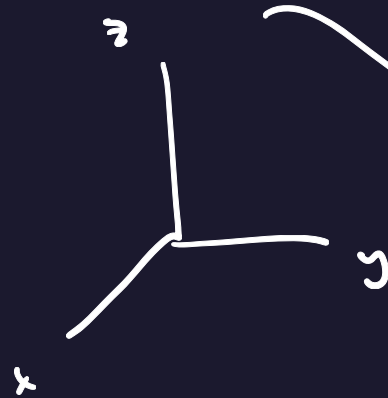
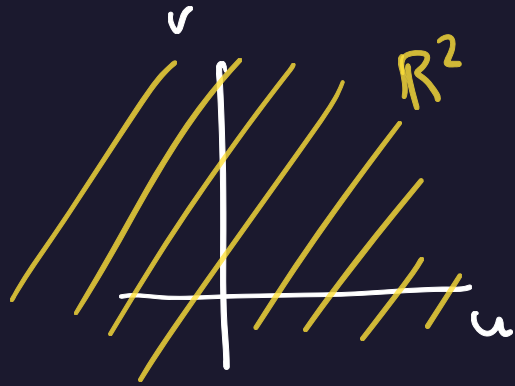


Example: Identify and sketch the surface with vector equation

$$r(u, v) = (1, u, u + v) = 1 \cdot i + u \cdot j + (u + v) \cdot k.$$



Example: Identify and sketch the surface with vector equation
 $r(u, v) = (u, u, u + v) = u \cdot i + u \cdot j + (u + v) \cdot k.$



$$S = \begin{cases} x = u \\ y = u \\ z = u + v \end{cases}$$

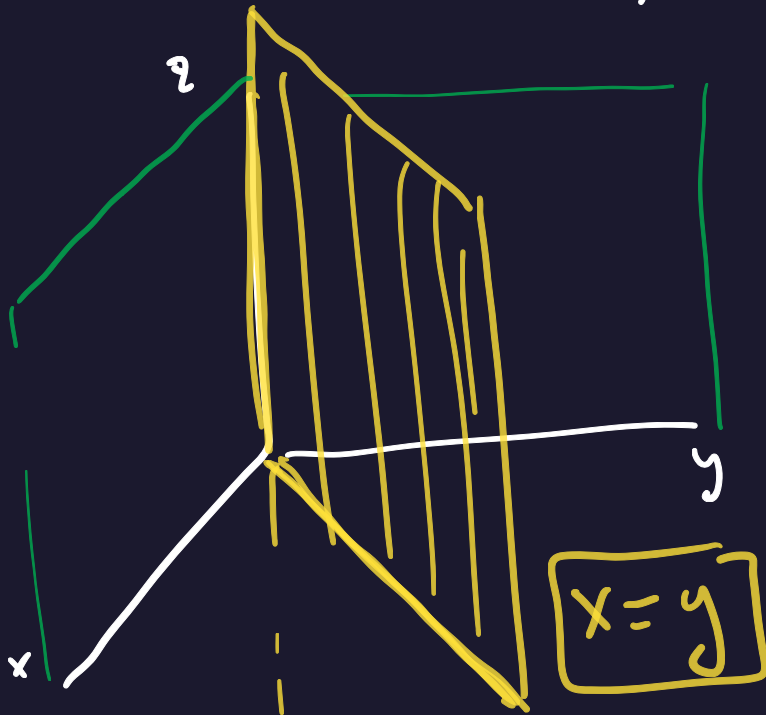
$$\cap \begin{cases} x = 0 \end{cases}$$

yz-plane

$$x = u = 0 = y \quad z = v$$

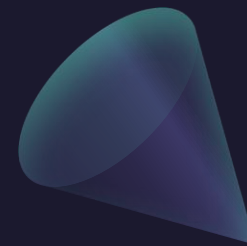
$$S \cap \{y = 0\} \leadsto \begin{cases} y = 0 = u = x \\ z = v \end{cases}$$

$$S \cap \{z = 0\} \leadsto \begin{cases} u + v = 0 \\ (u, u, 0) \end{cases} \left. \begin{array}{l} x = y \\ \text{xy plane} \end{array} \right\}$$

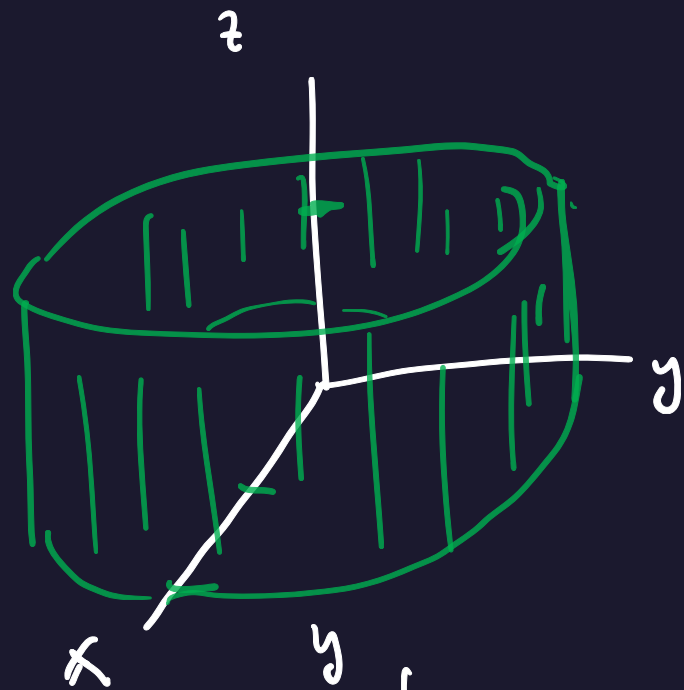


Example: Identify and sketch the surface with vector equation

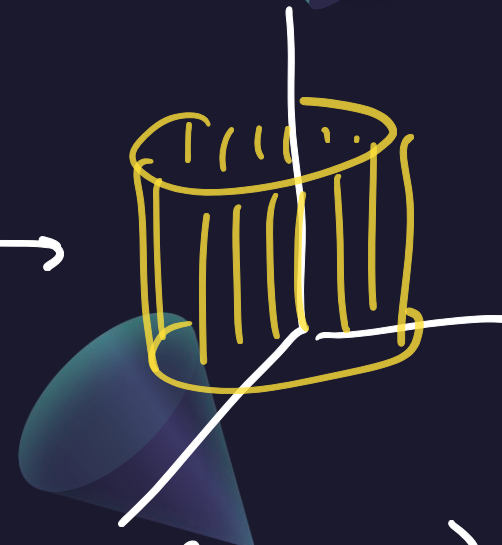
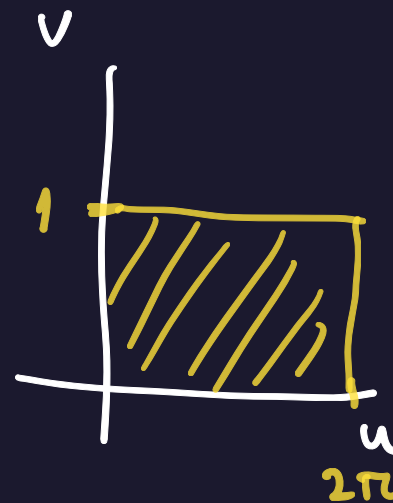
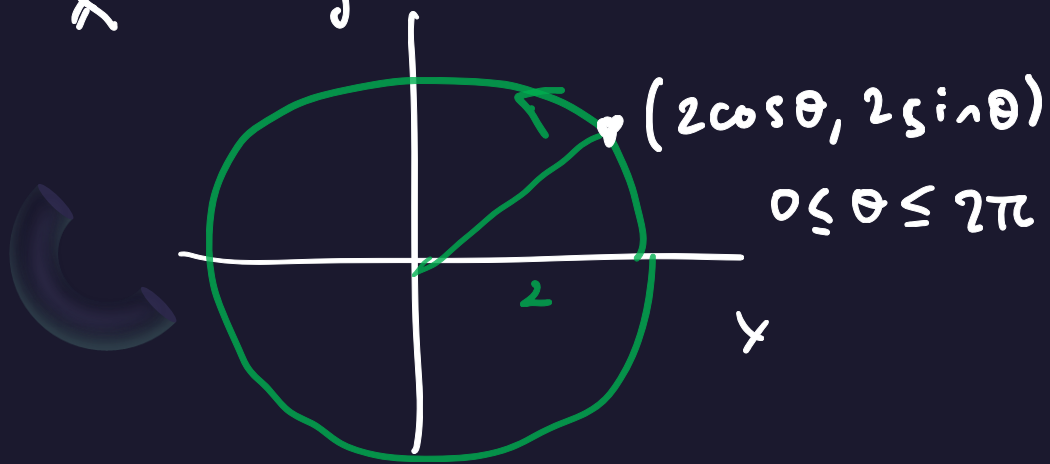
$$r(u, v) = (u, u, u + v) = u \cdot i + u \cdot j + (u + v) \cdot k.$$



Example: Find a parametric equation for the segment of a cylinder
 $x^2 + y^2 = 4, \quad 0 \leq z \leq 1.$

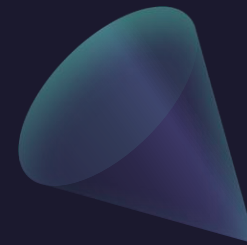
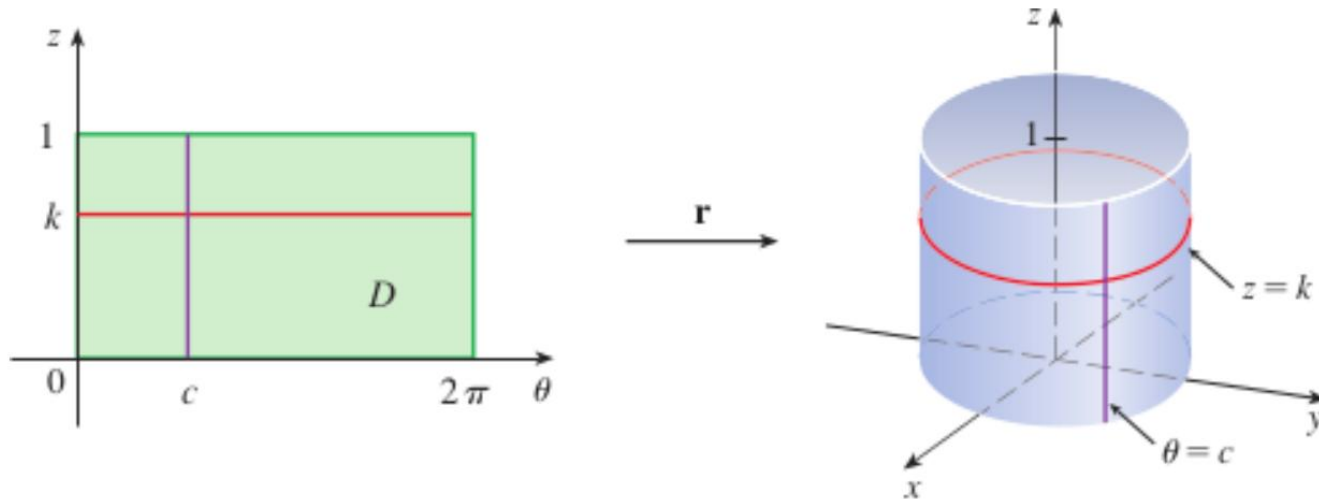
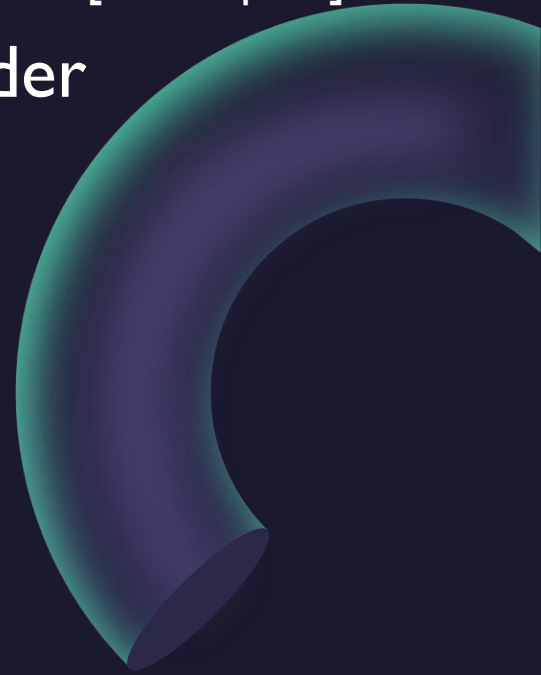


$$\begin{cases} x = 2\cos u \\ y = 2\sin u \\ z = v \end{cases} \quad \text{with} \quad \begin{aligned} 0 &\leq u \leq 2\pi \\ 0 &\leq v \leq 1 \end{aligned}$$



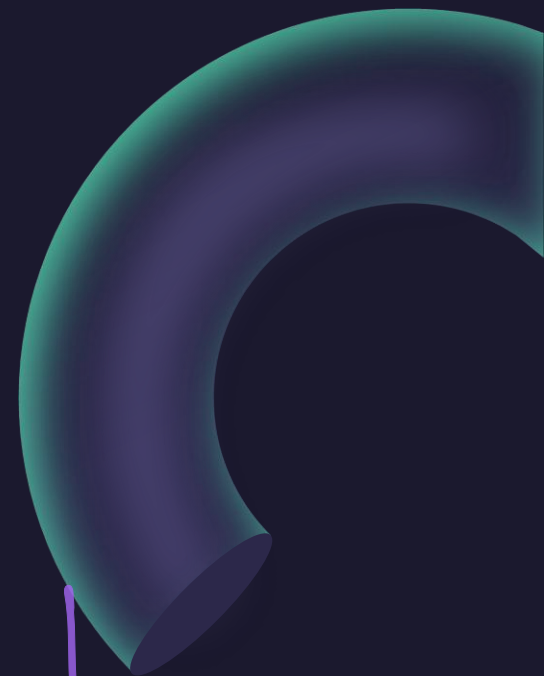
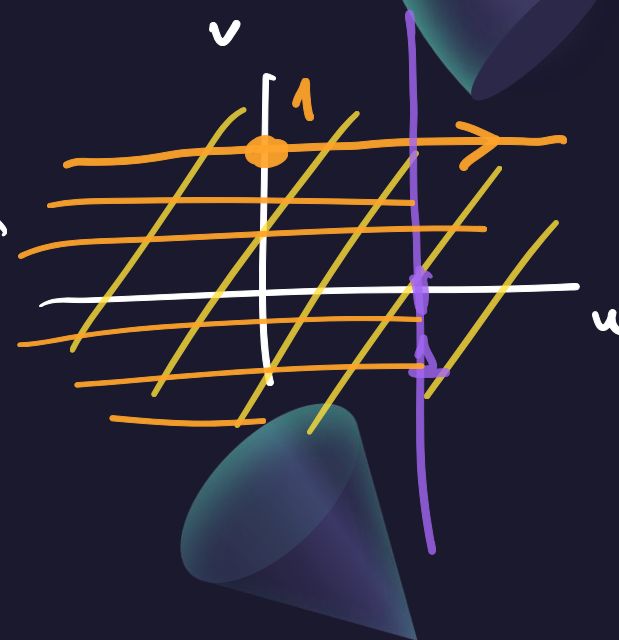
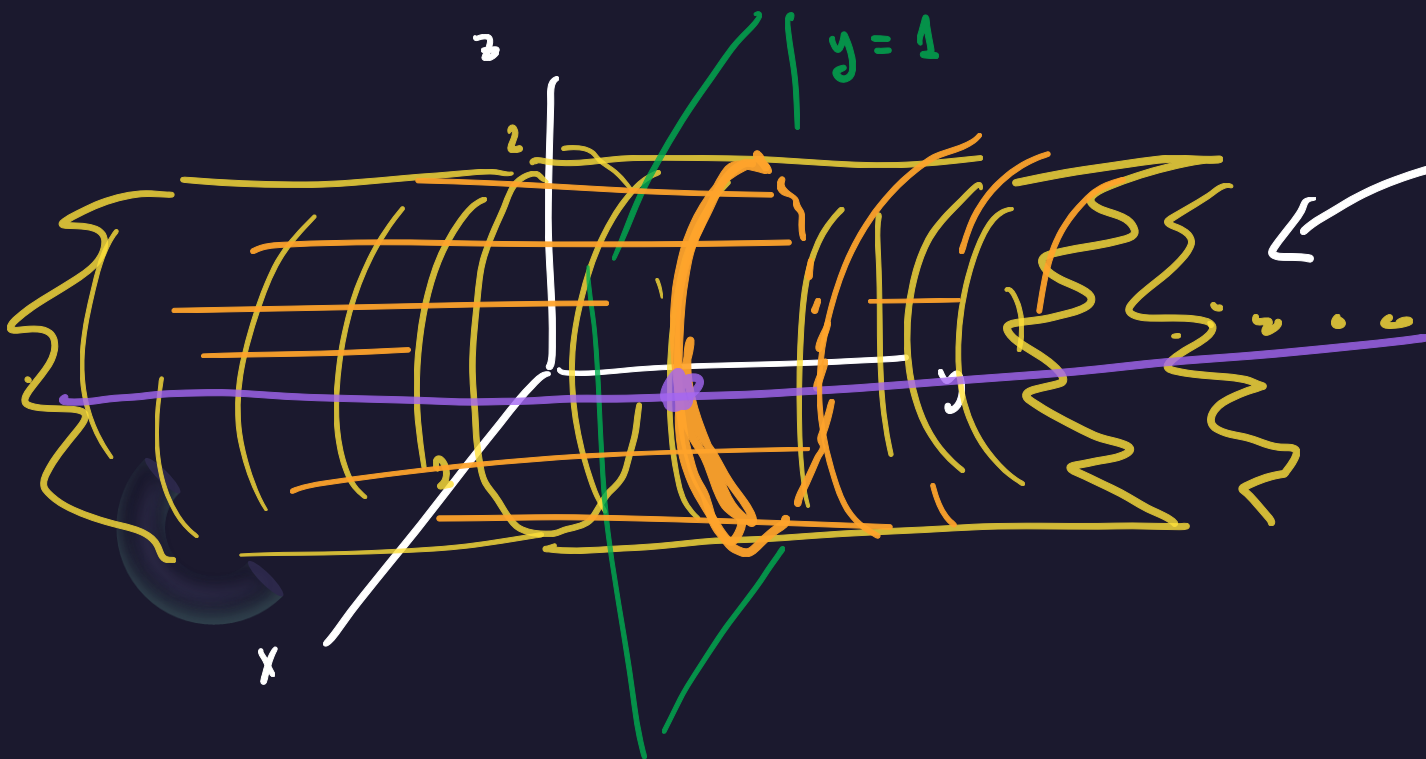
$$(u, v) \mapsto \mathbf{r}(u, v) = (2\cos u, 2\sin u, v)$$

Example: Find a parametric equation for the segment of a cylinder
 $x^2 + y^2 = 4, \quad 0 \leq z \leq 1.$

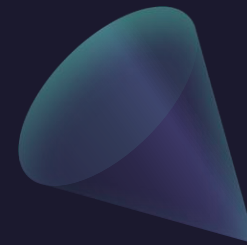
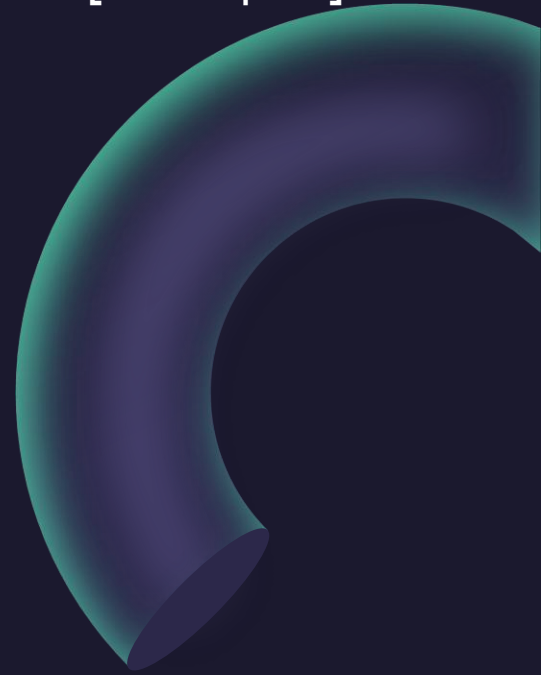


Example: Identify and sketch the surface with vector equation
 $r(u, v) = (2\cos(u), v, 2\sin(u))$

$$\begin{cases} x = 2\cos u \\ y = v \\ z = 2\sin u \end{cases} \quad \begin{matrix} \nearrow \\ \searrow \end{matrix} \quad x^2 + z^2 = 4$$



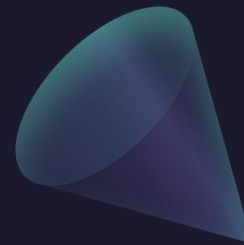
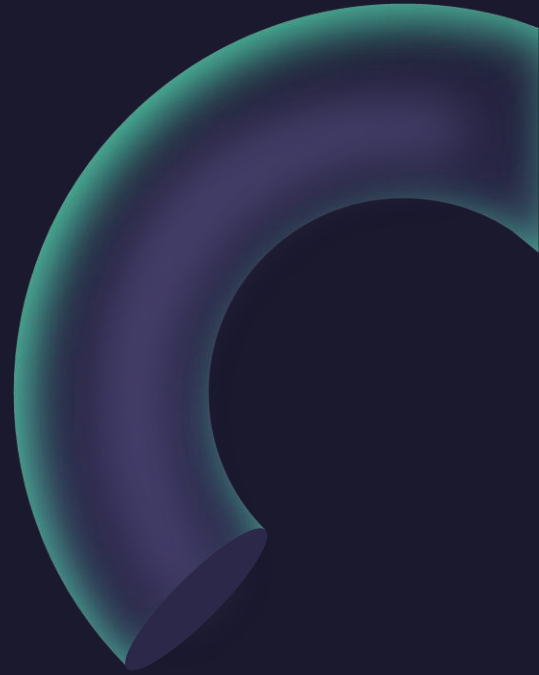
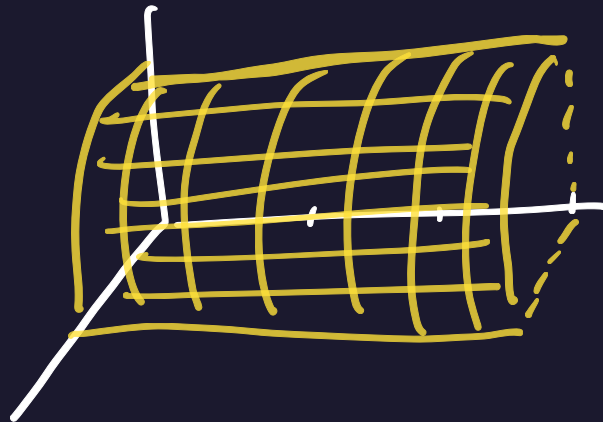
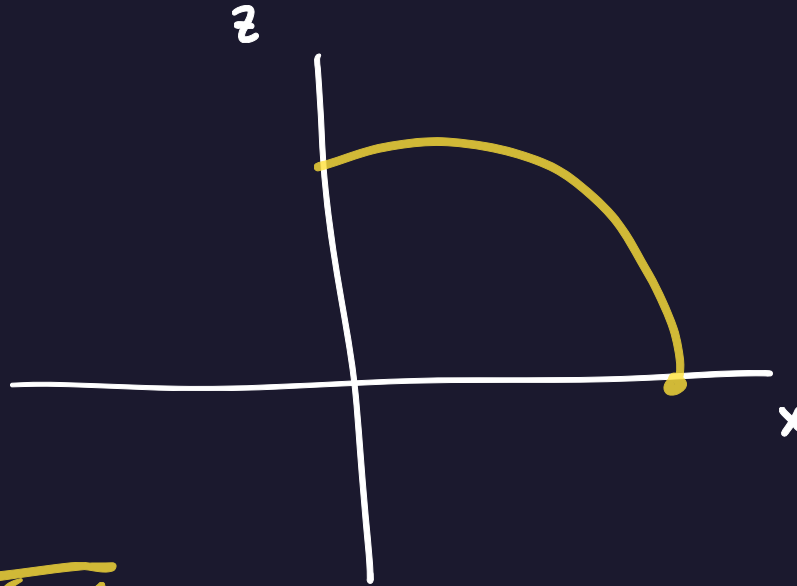
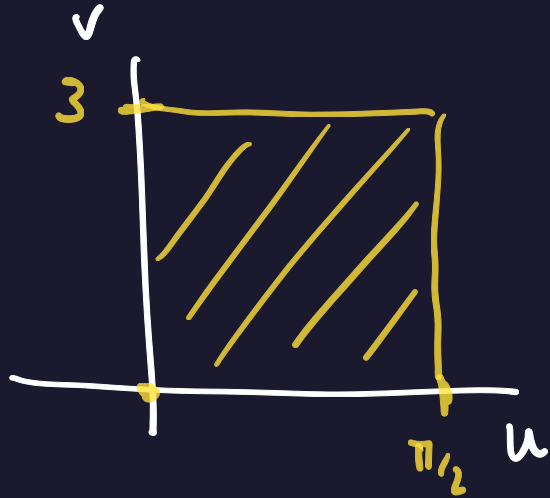
Example: Identify and sketch the surface with vector equation
$$r(u, v) = (2\cos(u), v, 2\sin(u))$$



Example: Identify and sketch the surface with vector equation

$$\mathbf{r}(u, v) = (2\cos(u), v, 2\sin(u))$$

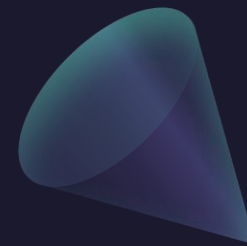
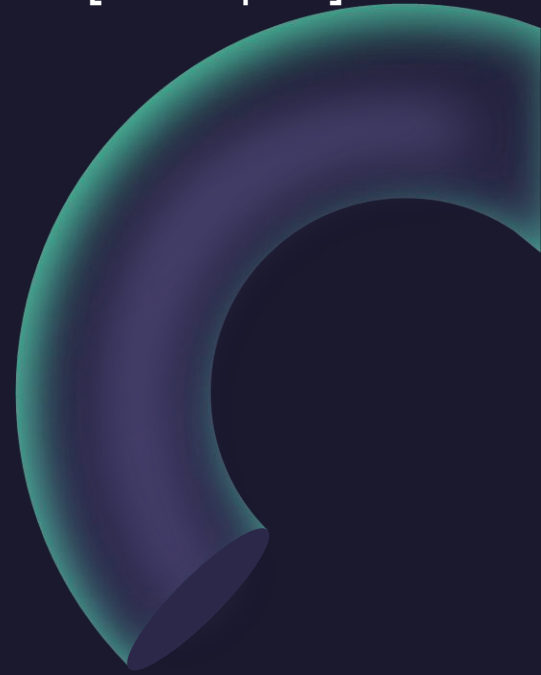
where $0 \leq u \leq \frac{\pi}{2}$, $0 \leq v \leq 3$.



Example: Identify and sketch the surface with vector equation

$$r(u, v) = (2\cos(u), v, 2\sin(u))$$

where $0 \leq u \leq \frac{\pi}{2}$, $0 \leq v \leq 3$.



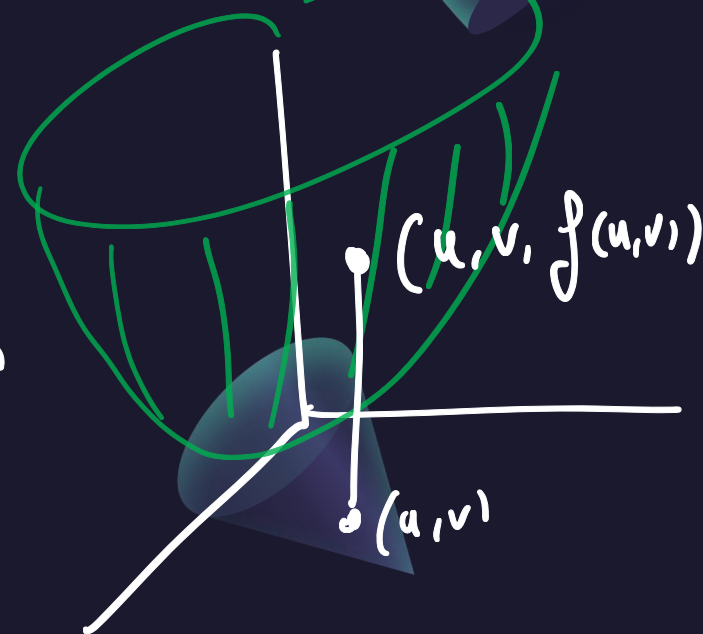
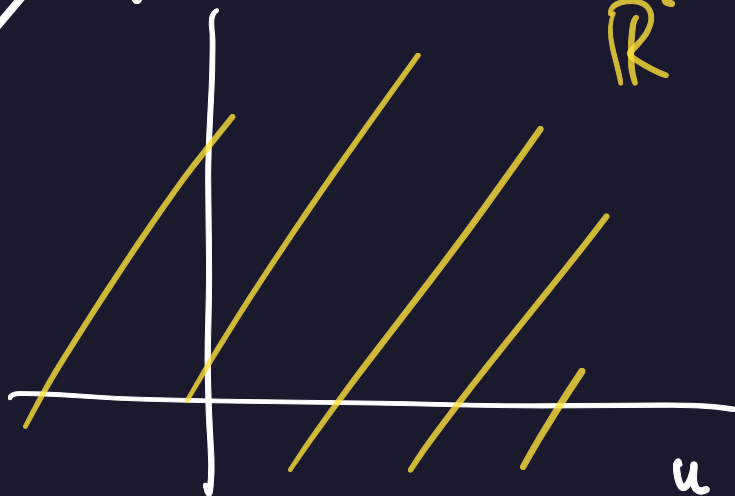
Example: Find a parametric equation for the elliptic paraboloid
 $z = x^2 + 2y^2$.

$$z = f(x, y)$$

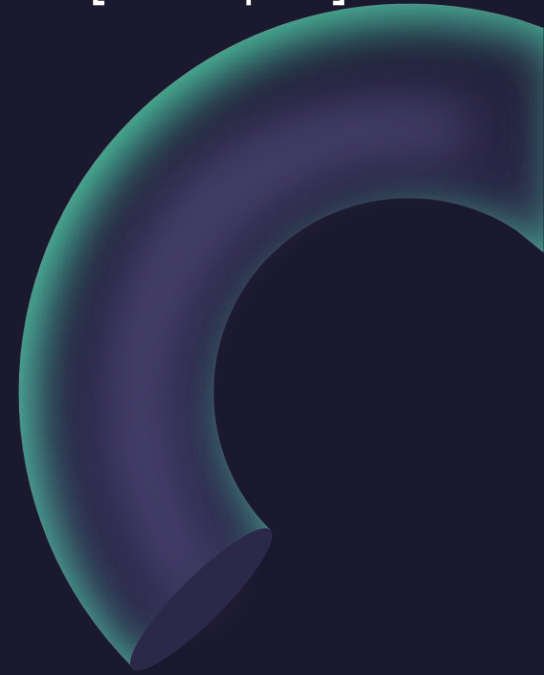
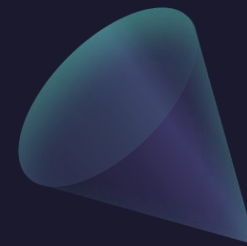
$$\mathbf{r}(u, v) = (u, v, u^2 + 2v^2)$$

$$\mathbf{r}(u, v) = (u, v, f(u, v))$$

$\mathbb{R}^2$



Example: Find a parametric equation for the elliptic paraboloid
 $z = x^2 + 2y^2$.



Parametric surfaces can be complicated!

$$(X(u,v), Y(u,v), Z(u,v))$$

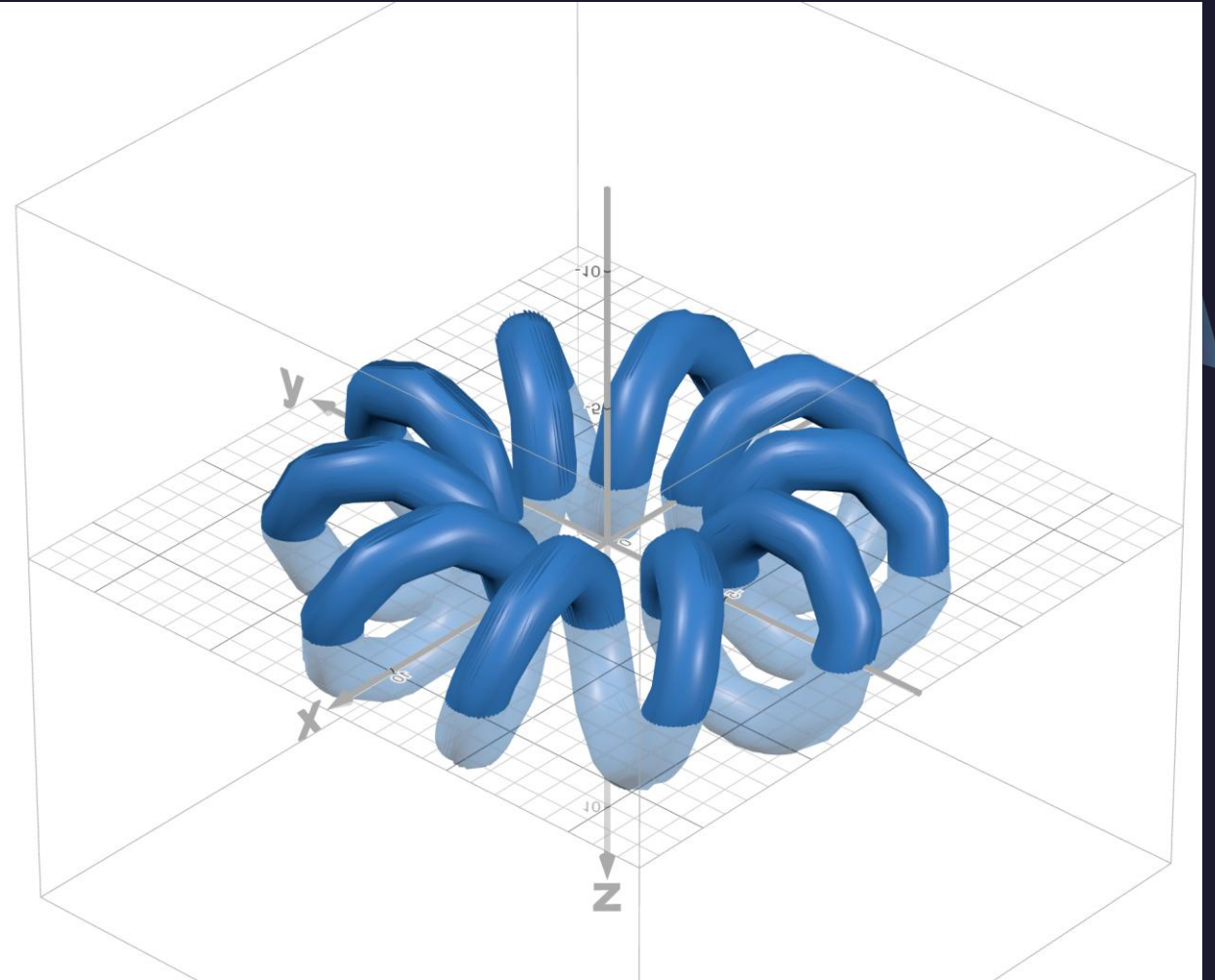
$$0 \leq u \leq 1$$

$$0 \leq v \leq 1$$

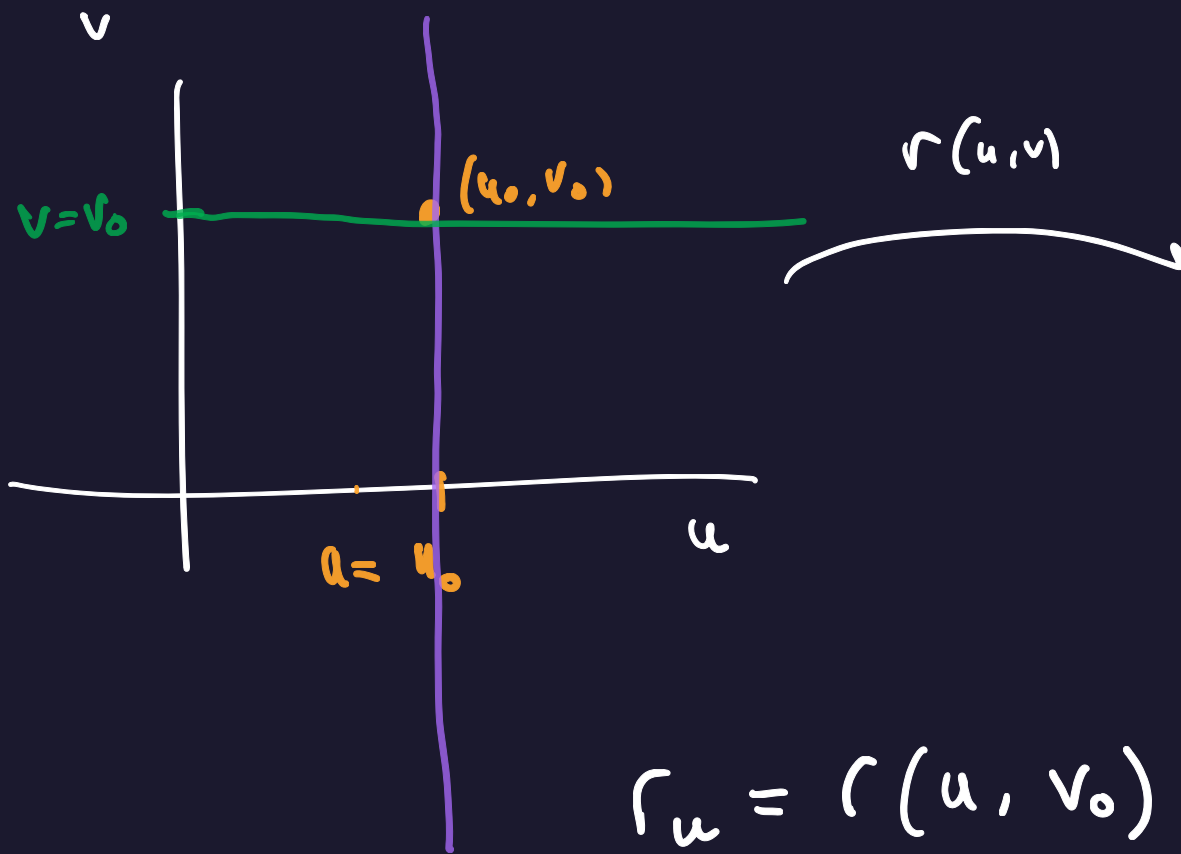
$$X(u,v) = \cos(\tau v) \cos(\tau u) + (\cos(10\tau v)(\sin(\tau u) + 3) + 7)(-\sin(\tau v))$$

$$Y(u,v) = \sin(\tau v) \cos(\tau u) + (\cos(10\tau v)(\sin(\tau u) + 3) + 7)(\cos(\tau v))$$

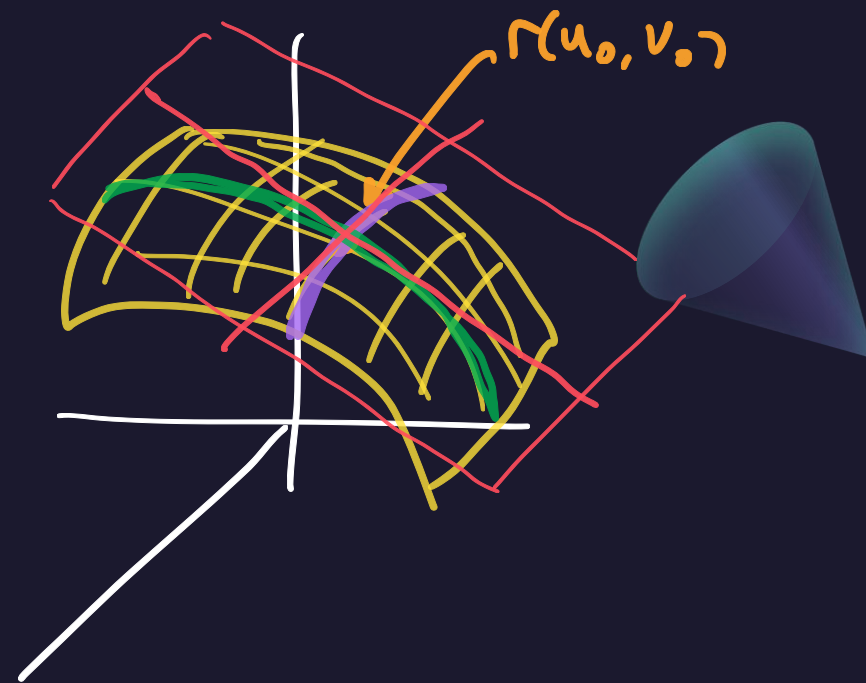
$$Z(u,v) = (\sin(10\tau v))(\sin(\tau u) + 3)$$



Grid curves on parametric surfaces



$$\begin{aligned}r_u &= r(u, v_0) \\ r_v &= r(u_0, v)\end{aligned}$$



Example: Sketch the “grid curves” with constant v

$$r(u, v) = ((2 + \sin(v)) \cos(u), (2 + \sin(v)) \sin(u), u + \cos(v))$$

$$v = v_0$$

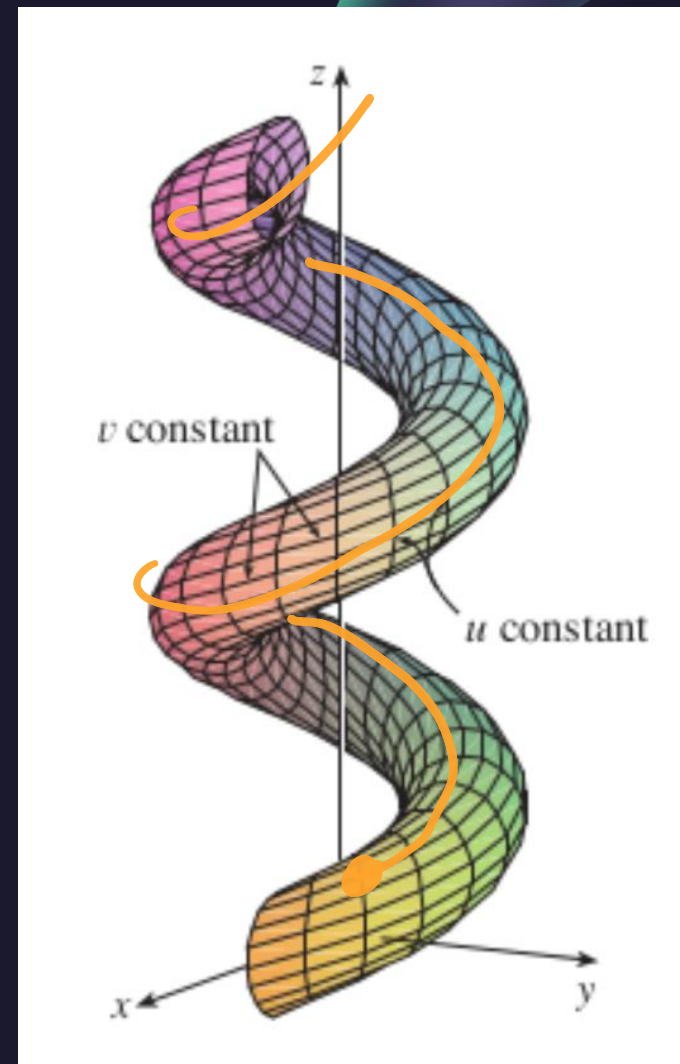
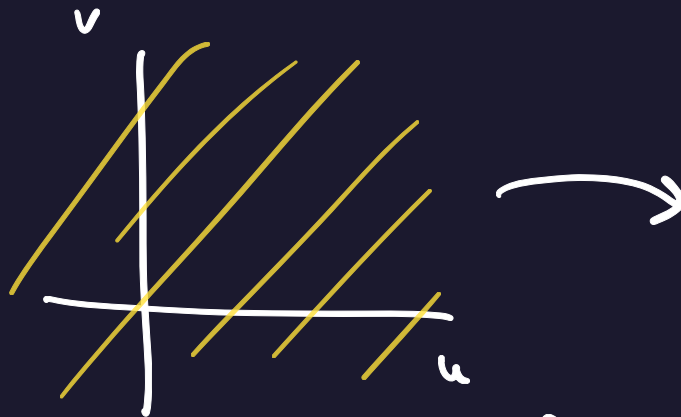
$$\Gamma(u, v_0)$$

$$= ((2 + \sin v_0) \cdot \cos u, (2 + \sin v_0) \sin u, u + \cos(v_0))$$

$$(2 + \sin v_0 = k > 0) \quad \cos v_0 = c$$

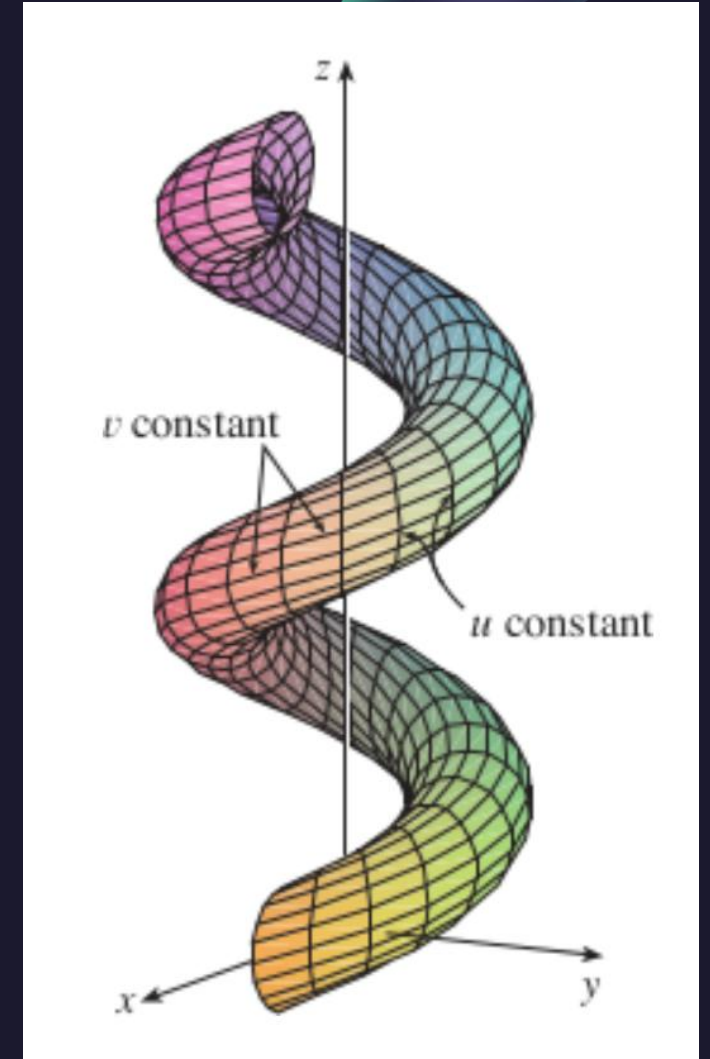
$$= (\underbrace{k \cos u, k \sin u}_{\text{circle radius } k}, \underbrace{u + c}_{\text{}})$$

circle radius k

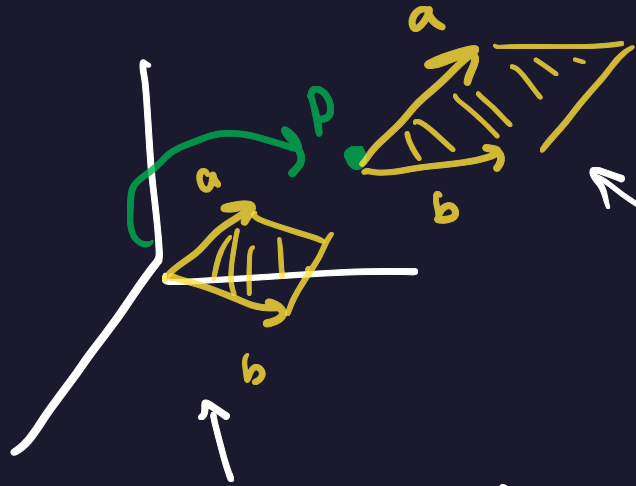


Example: Sketch the “grid curves” with constant v

$$r(u, v) = ((2 + \sin(v)) \cos(u), (2 + \sin(v)) \sin(u), u + \cos(v))$$



Example: Find a vector equation for the plane that passes through a point P and that contains two non-parallel vectors a and b .



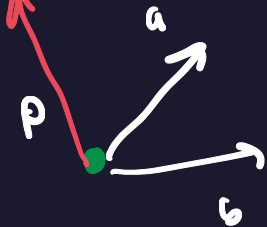
$$P = (x_0, y_0, z_0)$$

$$\vec{a} = (a_1, a_2, a_3)$$

$$\vec{b} = (b_1, b_2, b_3)$$

$$r(u, v) = u \cdot \vec{a} + v \cdot \vec{b}$$

$$\vec{a} \times \vec{b} = (c_1, c_2, c_3)$$



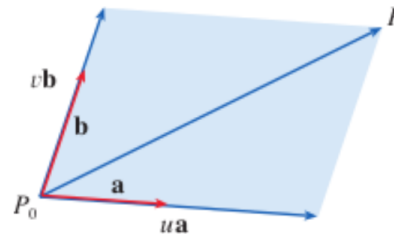
$$r(u, v) = P + u \cdot \vec{a} + v \cdot \vec{b}$$

$$= \begin{cases} x_0 + u a_1 + v b_1 \\ y_0 + u a_2 + v b_2 \\ z_0 + u a_3 + v b_3 \end{cases}$$

$$c_1(x - x_0) + c_2(y - y_0) + c_3(z - z_0) = 0$$

Example: Find a vector equation for the plane that passes through a point P and that contains two non-parallel vectors $\mathbf{a}$ and $\mathbf{b}$.

Figure 6



If we write $\mathbf{r} = \langle x, y, z \rangle$, $\mathbf{r}_0 = \langle x_0, y_0, z_0 \rangle$, $\mathbf{a} = \langle a_1, a_2, a_3 \rangle$, and $\mathbf{b} = \langle b_1, b_2, b_3 \rangle$, then we can write the parametric equations of the plane through the point (x_0, y_0, z_0) as follows:

$$x = x_0 + ua_1 + vb_1 \quad y = y_0 + ua_2 + vb_2 \quad z = z_0 + ua_3 + vb_3$$

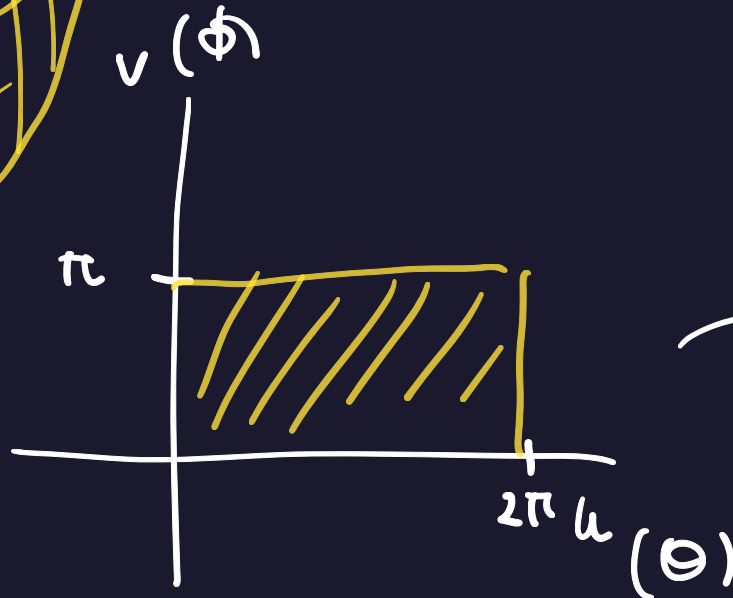
Example: Find a parametric equation for a sphere centered at $(0,0,0)$ of radius R .

surface of!

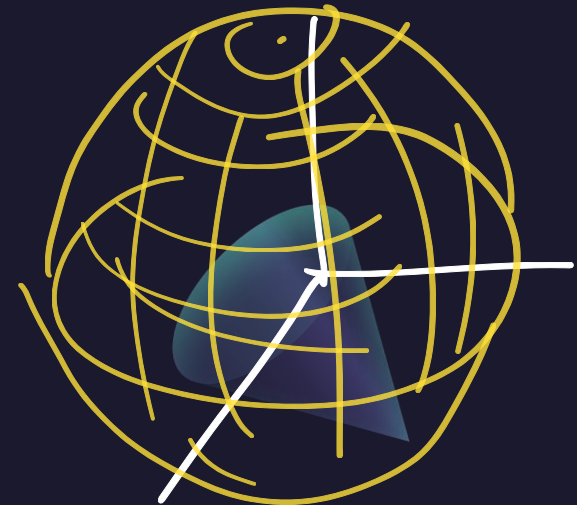


$$\left\{ \begin{array}{l} \rho = R \\ 0 \leq \theta \leq 2\pi \\ 0 \leq \phi \leq \pi \end{array} \right.$$

$$\left\{ \begin{array}{l} x = R \sin \phi \cos \theta \\ y = R \sin \phi \sin \theta \\ z = R \cos \phi \end{array} \right.$$

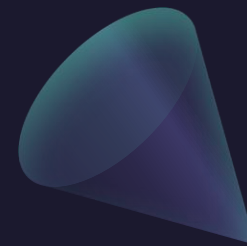
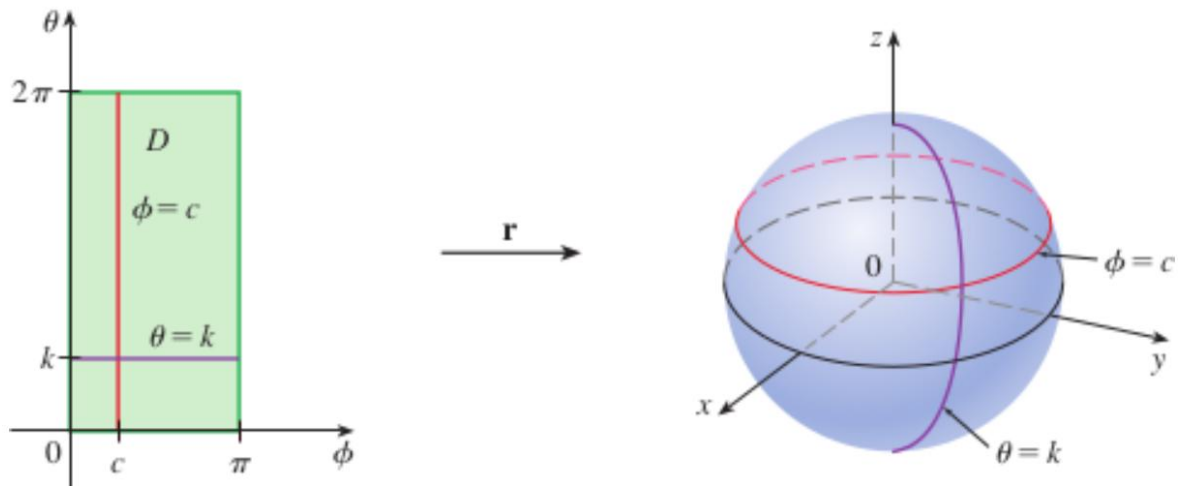


$$\vec{r}_R(u, v)$$



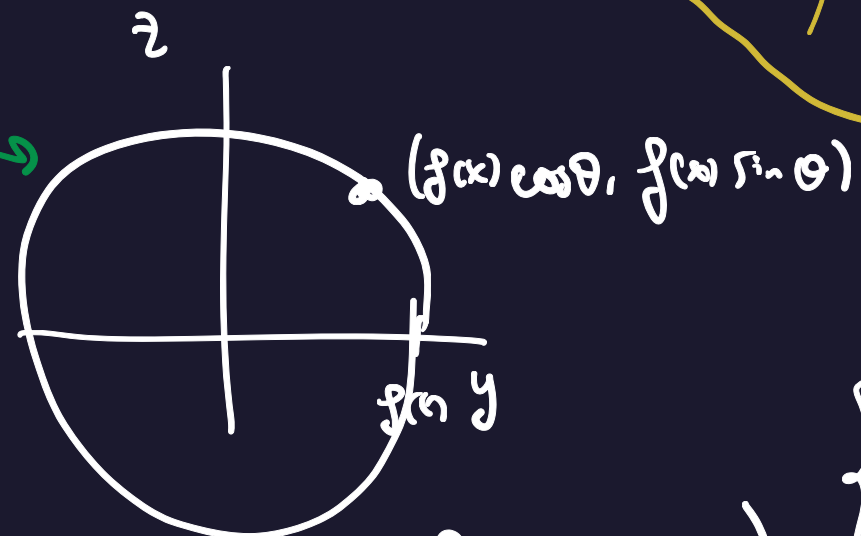
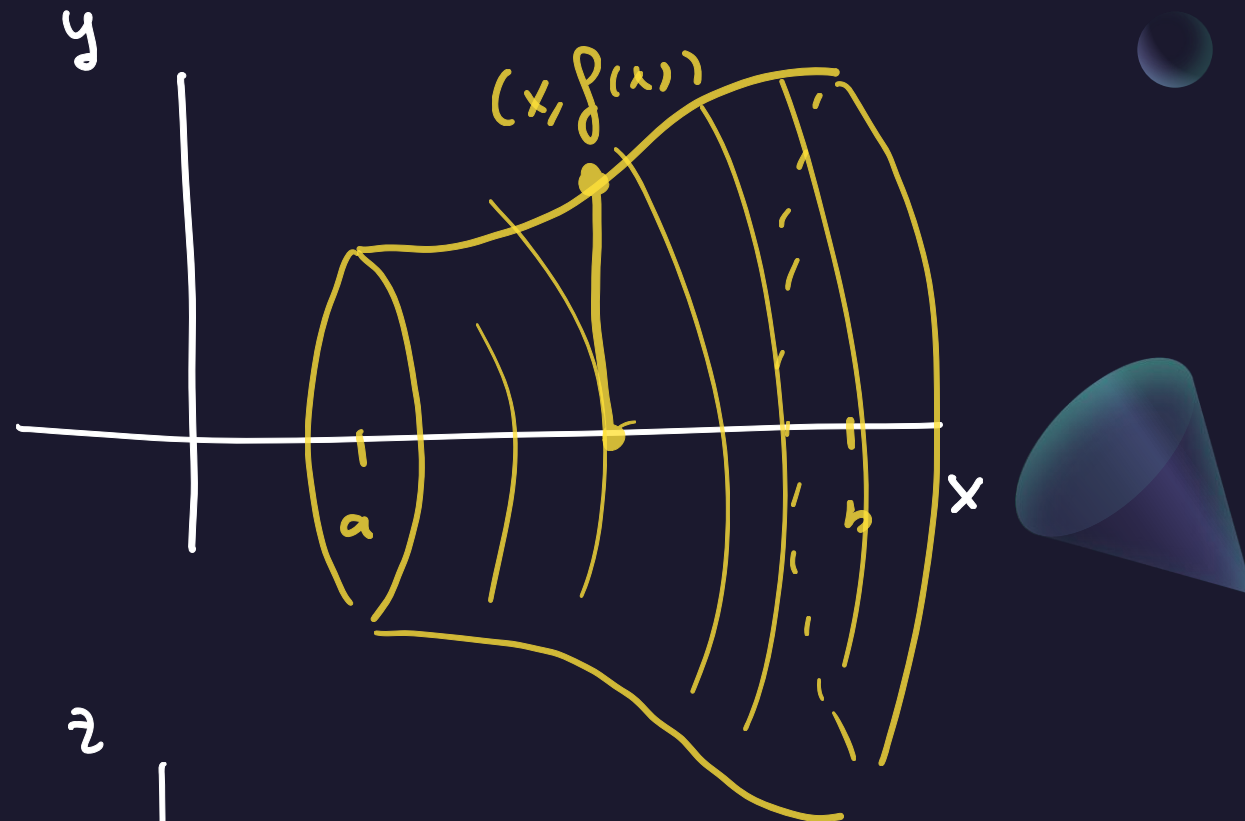
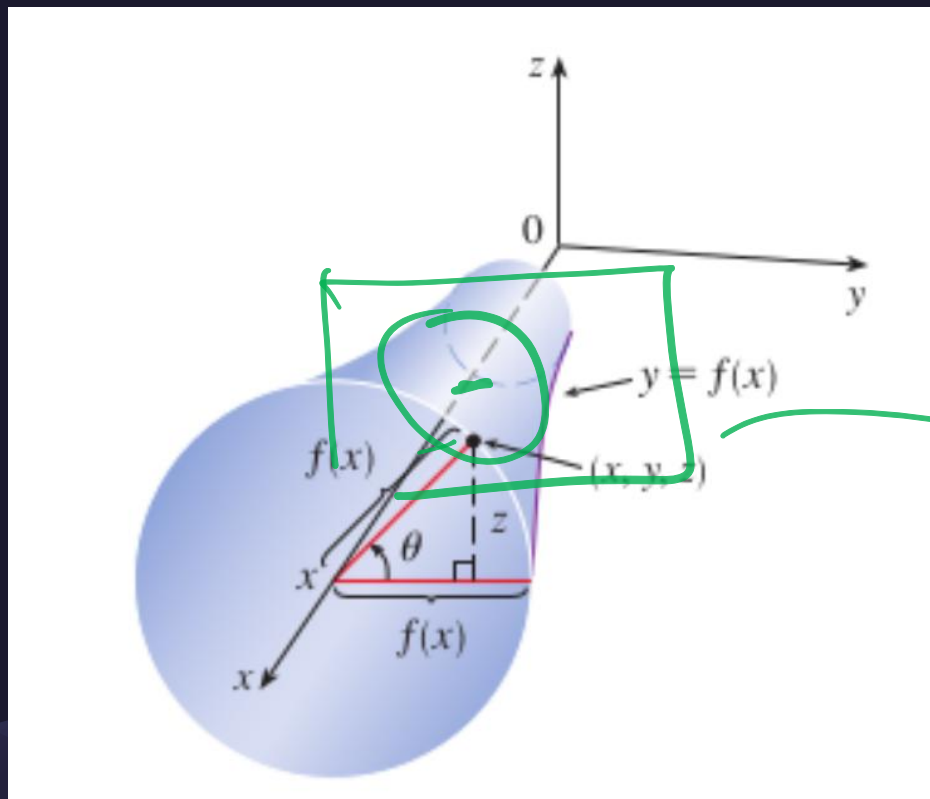
Example: Find a parametric equation for a sphere centered at $(0,0,0)$ of radius R .

$$x = a \sin \phi \cos \theta \quad y = a \sin \phi \sin \theta \quad z = a \cos \phi$$



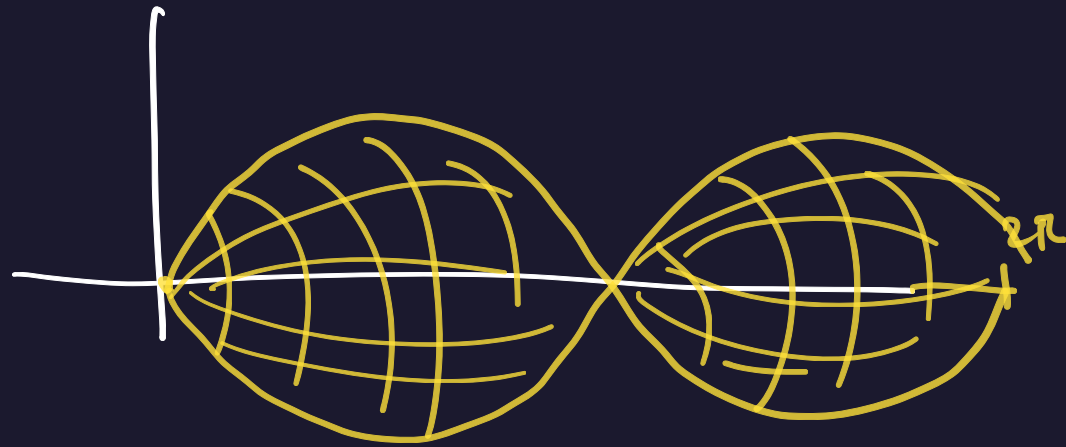
Surfaces of Revolution

$$x = x \quad y = f(x) \cos \theta \quad z = f(x) \sin \theta$$



$$\mathbf{r}(u, v) = (u, f(u) \cos v, f(u) \sin v) \quad \begin{cases} a \leq u \leq b \\ 0 \leq v \leq 2\pi \end{cases}$$

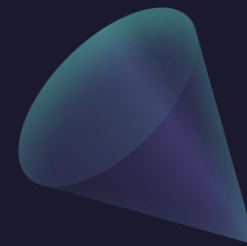
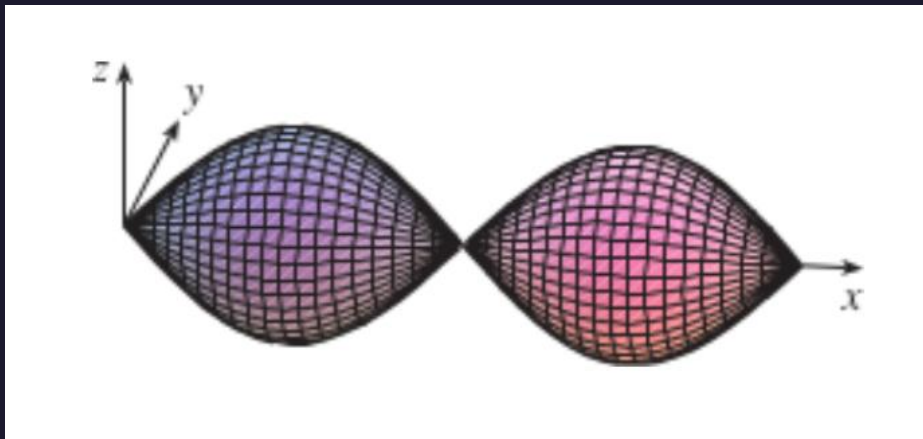
Example: Find a parametric equation for the surface that is obtained by rotating the curve $y = \sin(x)$, $0 \leq x \leq 2\pi$, about the x-axis.



$$r(u, v) = (u, \sin(u) \cos(v), \sin(u) \sin(v))$$

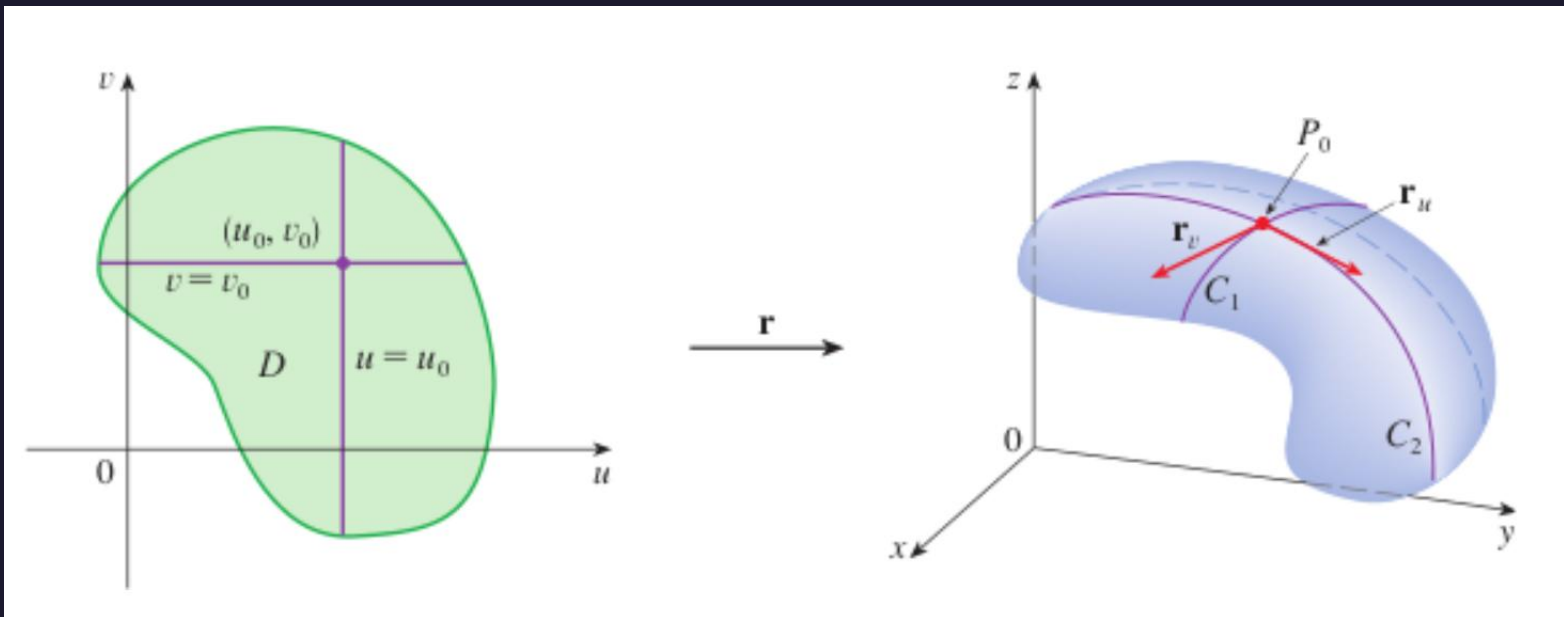
$$\begin{cases} 0 \leq u \leq 2\pi \\ 0 \leq v \leq 2\pi \end{cases}$$

Example: Find a parametric equation for the surface that is obtained by rotating the curve $y = \sin(x)$, $0 \leq x \leq 2\pi$, about the x -axis.



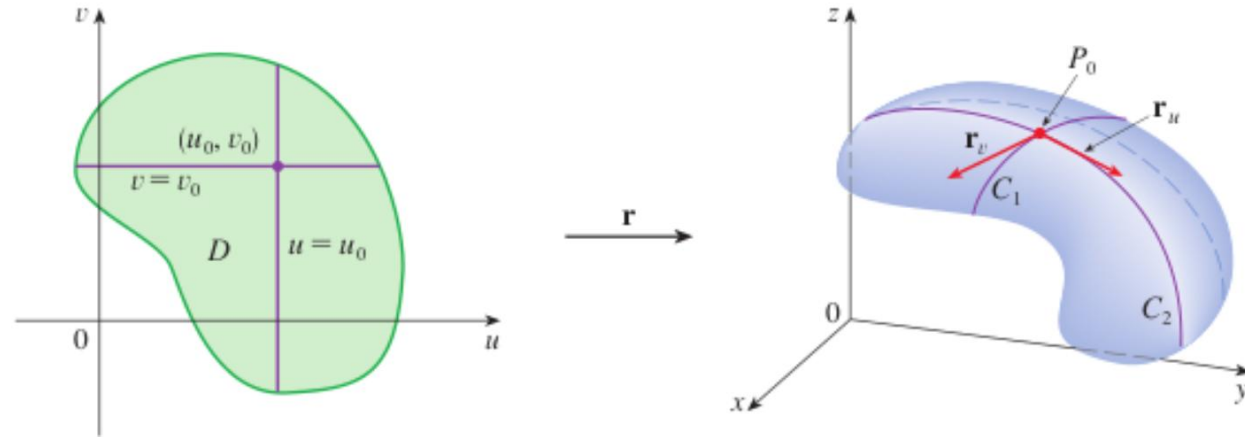
Tangent Planes

$$\mathbf{r}(u, v) = x(u, v) \mathbf{i} + y(u, v) \mathbf{j} + z(u, v) \mathbf{k}$$



Tangent Planes

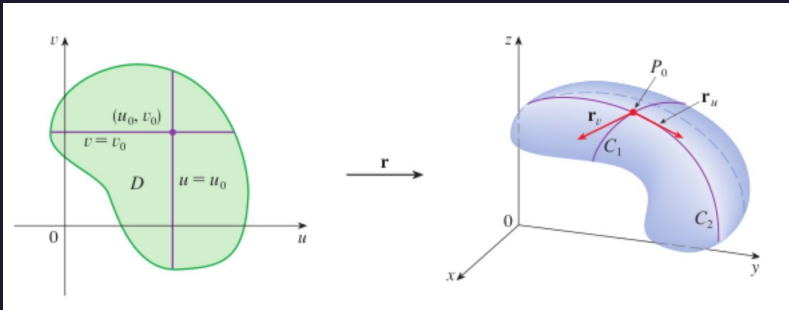
$$\mathbf{r}(u, v) = x(u, v) \mathbf{i} + y(u, v) \mathbf{j} + z(u, v) \mathbf{k}$$



$$\mathbf{r}_u = \frac{\partial x}{\partial u}(u_0, v_0) \mathbf{i} + \frac{\partial y}{\partial u}(u_0, v_0) \mathbf{j} + \frac{\partial z}{\partial u}(u_0, v_0) \mathbf{k}$$

$$\mathbf{r}_v = \frac{\partial x}{\partial v}(u_0, v_0) \mathbf{i} + \frac{\partial y}{\partial v}(u_0, v_0) \mathbf{j} + \frac{\partial z}{\partial v}(u_0, v_0) \mathbf{k}$$

Tangent Planes



$$\mathbf{r}(u, v) = x(u, v) \mathbf{i} + y(u, v) \mathbf{j} + z(u, v) \mathbf{k}$$

$$\mathbf{r}_u = \frac{\partial x}{\partial u}(u_0, v_0) \mathbf{i} + \frac{\partial y}{\partial u}(u_0, v_0) \mathbf{j} + \frac{\partial z}{\partial u}(u_0, v_0) \mathbf{k}$$

$$\mathbf{r}_v = \frac{\partial x}{\partial v}(u_0, v_0) \mathbf{i} + \frac{\partial y}{\partial v}(u_0, v_0) \mathbf{j} + \frac{\partial z}{\partial v}(u_0, v_0) \mathbf{k}$$

If $\mathbf{r}_u \times \mathbf{r}_v$ is not $\mathbf{0}$, then the surface S is called **smooth** (it has no “corners”). For a smooth surface, the **tangent plane** is the plane that contains the tangent vectors $\mathbf{r}_u$ and $\mathbf{r}_v$, and the vector $\mathbf{r}_u \times \mathbf{r}_v$ is a normal vector to the tangent plane.

Example: Find the equation of the tangent plane to the surface with parametric equations $x = u^2, y = v^2, z = u + 2v$ at the point $P=(1,1,3)$.

$$\mathbf{r}(u,v) = (u^2, v^2, u+2v), \quad P = (1, 1, 3)$$

- $u^2=1, v^2=1, u+2v=3 \rightarrow (u,v) = (1,1)$

- $\mathbf{r}_u = (2u, 0, 1) \rightarrow \mathbf{r}_u(1,1) = (2, 0, 1)$

- $\mathbf{r}_v = (0, 2v, 2) \rightarrow \mathbf{r}_v(1,1) = (0, 2, 2)$

- $\mathbf{r}_u \times \mathbf{r}_v = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 2 & 0 & 1 \\ 0 & 2 & 2 \end{vmatrix} = (-2, 4, 4) \rightarrow \mathbf{n} = (-1, 2, 2)$

Example: Find the equation of the tangent plane to the surface with parametric equations $x = u^2, y = v^2, z = u + 2v$ at the point $P=(1,1,3)$.

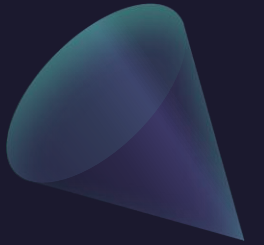
$$P = (1, 1, 3)$$

$$n = (-1, 2, 2)$$

$$P_{\text{plane}} : (-1) \cdot (x-1) + 2 \cdot (y-1) + 2 \cdot (z-3) = 0$$

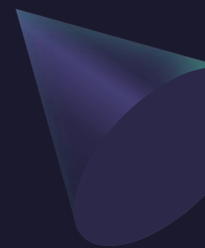
$$x + 2y - 2z = -3$$

Questions?





ALVARO: Start the recording!



“Calculus 3”

Multi-Variable Calculus

Instructor: Álvaro Lozano-Robledo

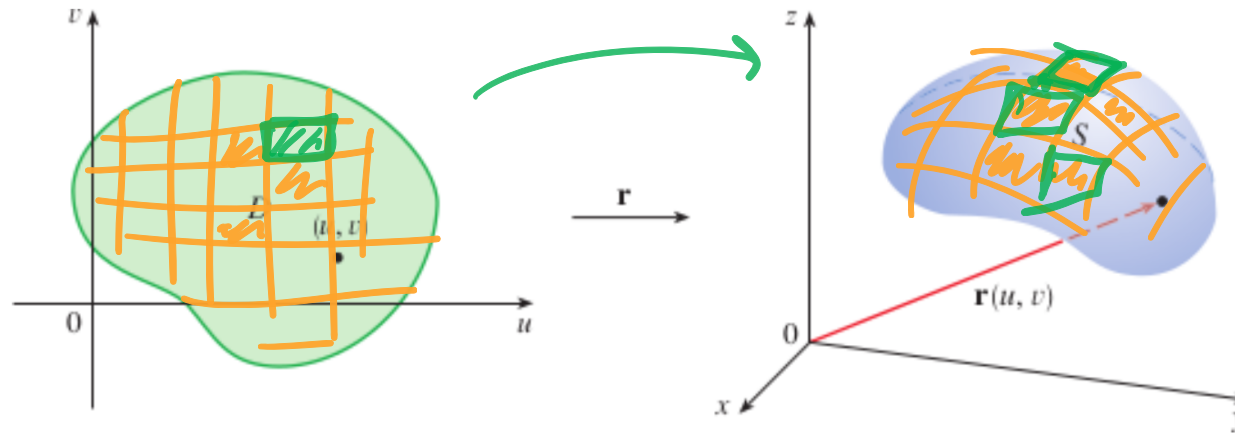
Surface Area



About Parametrized Surfaces and their Area

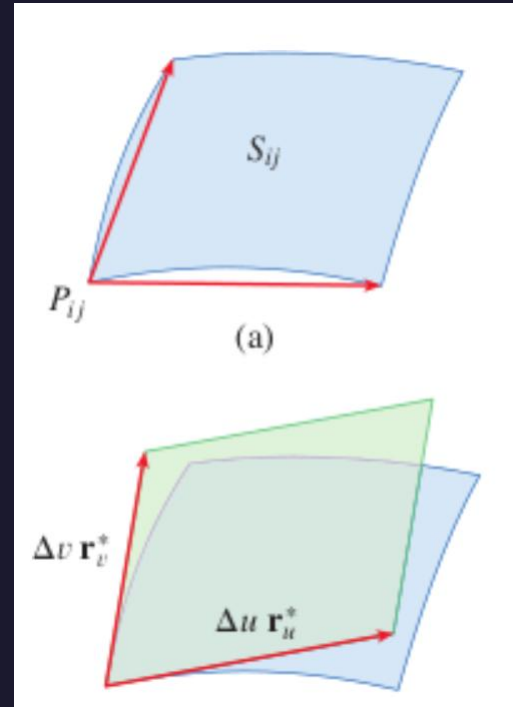
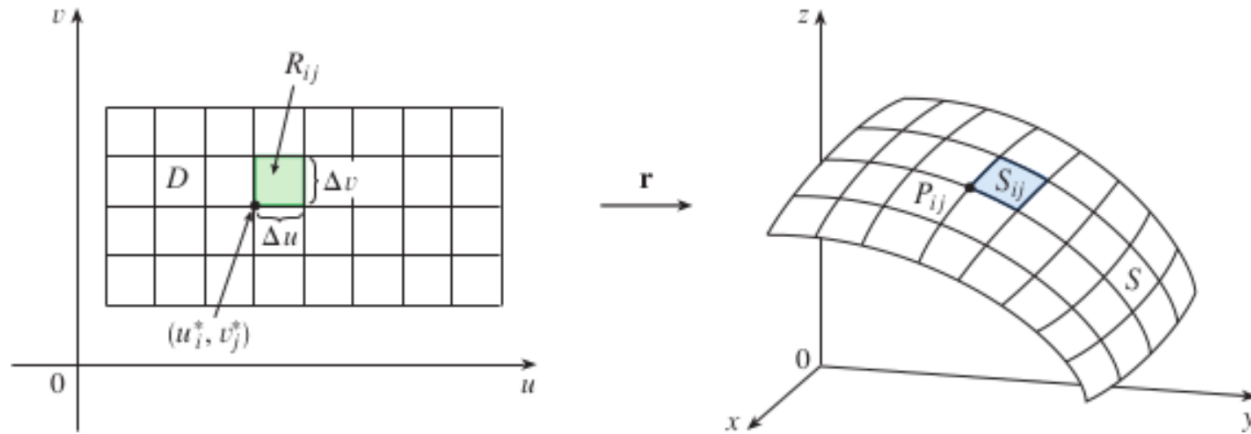
$$\mathbf{r}(u, v) = x(u, v) \mathbf{i} + y(u, v) \mathbf{j} + z(u, v) \mathbf{k}$$

A parametric surface



About Parametrized Surfaces and their Area

The image of the subrectangle R_{ij} is the patch S_{ij} .



$$|(\Delta u \mathbf{r}_u^*) \times (\Delta v \mathbf{r}_v^*)| = |\mathbf{r}_u^* \times \mathbf{r}_v^*| \Delta u \Delta v$$

$$\sum_{i=1}^m \sum_{j=1}^n |\mathbf{r}_u^* \times \mathbf{r}_v^*| \Delta u \Delta v$$

About Parametrized Surfaces and their Area

If a smooth parametric surface S is given by the equation

$$\mathbf{r}(u, v) = x(u, v) \mathbf{i} + y(u, v) \mathbf{j} + z(u, v) \mathbf{k} \quad (u, v) \in D$$

and S is covered just once as (u, v) ranges throughout the parameter domain D , then the **surface area** of S is

$$A(S) = \iint_D |\mathbf{r}_u \times \mathbf{r}_v| dA$$

where

$$\mathbf{r}_u = \frac{\partial x}{\partial u} \mathbf{i} + \frac{\partial y}{\partial u} \mathbf{j} + \frac{\partial z}{\partial u} \mathbf{k} \quad \mathbf{r}_v = \frac{\partial x}{\partial v} \mathbf{i} + \frac{\partial y}{\partial v} \mathbf{j} + \frac{\partial z}{\partial v} \mathbf{k}$$

Example: Compute the surface area of the segment of a cylinder
 $x^2 + y^2 = 4, \quad 0 \leq z \leq 1.$

$$\mathbf{r}(u, v) = (2 \cos u, 2 \sin u, v)$$

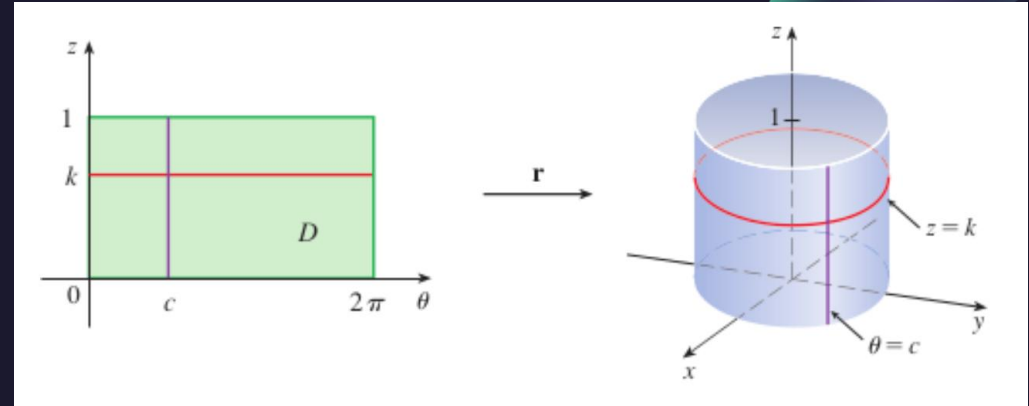
$$0 \leq u \leq 2\pi, \quad 0 \leq v \leq 1$$

$$\mathbf{r}_u = (-2 \sin u, 2 \cos u, 0)$$

$$\mathbf{r}_v = (0, 0, 1)$$

$$\mathbf{r}_u \times \mathbf{r}_v = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ -2 \sin u & 2 \cos u & 0 \\ 0 & 0 & 1 \end{vmatrix}$$

$$= (2 \cos u, 2 \sin u, 0), \quad |\mathbf{r}_u \times \mathbf{r}_v| = \sqrt{4 \cos^2 u + 4 \sin^2 u} = 2$$



$$S = 2\pi r \cdot h$$

$$= 2\pi \cdot 2 \cdot 1 = 4\pi$$

Example: Find a parametric equation for the segment of a cylinder

$$x^2 + y^2 = 4, \quad 0 \leq z \leq 1.$$

$$A(S) = \iint_D |\mathbf{r}_u \times \mathbf{r}_v| \, dA$$

$$= \int_0^{2\pi} \int_0^1 2 \, dv \, du = 2 \cdot \int_0^{2\pi} \int_0^1 1 \, dv \, du$$

$$= 2 \cdot 2\pi = 4\pi$$

Example: Compute the surface area of a sphere of radius R .

$$\mathbf{r}(\theta, \phi) = (R \sin \phi \cos \theta, R \sin \phi \sin \theta, R \cos \phi)$$

$$0 \leq \phi \leq \pi, \quad 0 \leq \theta \leq 2\pi$$

$$\mathbf{r}_\theta = (-R \sin \phi \sin \theta, R \sin \phi \cos \theta, 0)$$

$$\mathbf{r}_\phi = (R \cos \phi \cos \theta, R \cos \phi \sin \theta, -R \sin \phi)$$

$$|\mathbf{r}_\phi \times \mathbf{r}_\theta| = R^2 \sin \phi$$

$$\int_0^{2\pi} \int_0^\pi R^2 \sin \phi \, d\phi \, d\theta$$

$$= \boxed{4\pi R^2} = \left(\frac{4}{3}\pi R^3\right)'$$

$$x = a \sin \phi \cos \theta \quad y = a \sin \phi \sin \theta \quad z = a \cos \phi$$

Example: Compute the surface area of a sphere of radius R .

$$\begin{aligned}\mathbf{r}_\phi \times \mathbf{r}_\theta &= \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ \frac{\partial x}{\partial \phi} & \frac{\partial y}{\partial \phi} & \frac{\partial z}{\partial \phi} \\ \frac{\partial x}{\partial \theta} & \frac{\partial y}{\partial \theta} & \frac{\partial z}{\partial \theta} \end{vmatrix} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ a \cos \phi \cos \theta & a \cos \phi \sin \theta & -a \sin \phi \\ -a \sin \phi \sin \theta & a \sin \phi \cos \theta & 0 \end{vmatrix} \\ &= a^2 \sin^2 \phi \cos \theta \mathbf{i} + a^2 \sin^2 \phi \sin \theta \mathbf{j} + a^2 \sin \phi \cos \phi \mathbf{k}\end{aligned}$$

$$\begin{aligned}|\mathbf{r}_\phi \times \mathbf{r}_\theta| &= \sqrt{a^4 \sin^4 \phi \cos^2 \theta + a^4 \sin^4 \phi \sin^2 \theta + a^4 \sin^2 \phi \cos^2 \phi} \\ &= \sqrt{a^4 \sin^4 \phi + a^4 \sin^2 \phi \cos^2 \phi} = a^2 \sqrt{\sin^2 \phi} = a^2 \sin \phi\end{aligned}$$

Surfaces Area of a Graph $z = f(x,y)$

$$\mathbf{r}(u,v) = (u, v, f(u,v))$$

$$\mathbf{r}_u = \left(1, 0, \frac{\partial f}{\partial u} \right)$$

$$\mathbf{r}_v = \left(0, 1, \frac{\partial f}{\partial v} \right)$$

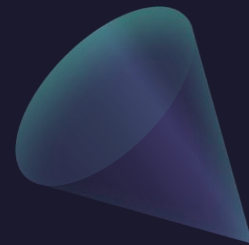
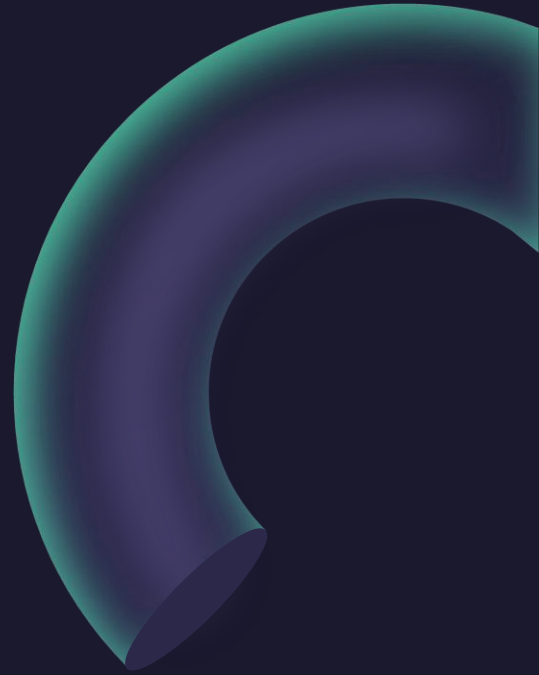
$$A = \iint_D \sqrt{1 + \frac{\partial f^2}{\partial u} + \frac{\partial f^2}{\partial v}} \, du \, dv$$

Surfaces Area of a Graph (and Length of a Graph)

$$A(S) = \iint_D \sqrt{1 + \left(\frac{\partial z}{\partial x}\right)^2 + \left(\frac{\partial z}{\partial y}\right)^2} dA$$

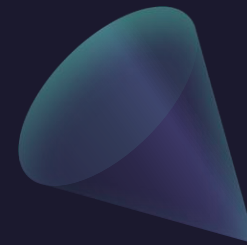
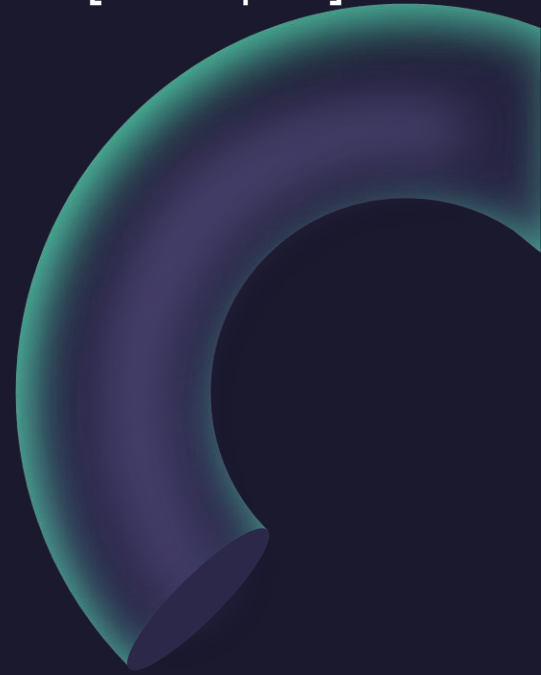
$$L = \int_a^b \sqrt{1 + \left(\frac{dy}{dx}\right)^2} dx$$

Example: Compute the surface area of the paraboloid $z = x^2 + y^2$ below the plane $z = 9$.

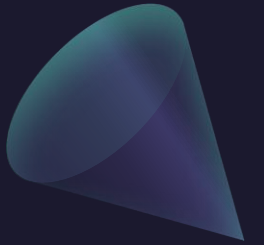


$$x = a \sin \phi \cos \theta \quad y = a \sin \phi \sin \theta \quad z = a \cos \phi$$

Example: Compute the surface area of the paraboloid $z = x^2 + y^2$ below the plane $z = 9$.



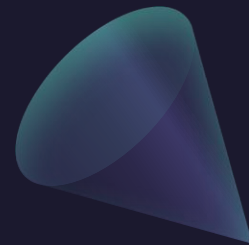
Surfaces Area of a Surface of Revolution



Surfaces Area of a Surface of Revolution

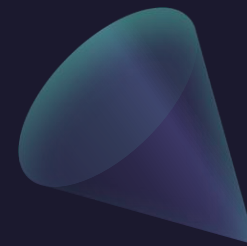
$$\begin{aligned} A &= \iint_D |\mathbf{r}_x \times \mathbf{r}_\theta| \, dA \\ &= \int_0^{2\pi} \int_a^b f(x) \sqrt{1 + [f'(x)]^2} \, dx \, d\theta \\ &= 2\pi \int_a^b f(x) \sqrt{1 + [f'(x)]^2} \, dx \end{aligned}$$

Example: Compute the surface area of the paraboloid $z = x^2 + y^2$ below the plane $z = 9$, as a surface of revolution.

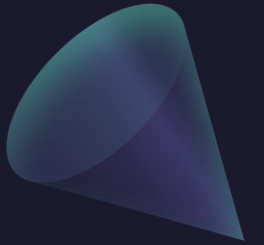


$$x = a \sin \phi \cos \theta \quad y = a \sin \phi \sin \theta \quad z = a \cos \phi$$

Example: Compute the surface area of the paraboloid $z = x^2 + y^2$ below the plane $z = 9$, as a surface of revolution.



Questions?



Thank you

Until next time.

